

Open *E. coli* Model

River Deep Mountain AI

Hossein Darvishi^{1*}, Nicolai Tarp¹, Daisy Allen², Richard Gooday³, Chris Dawson⁴

¹ Cognizant ² WRc ³ ADAS ⁴ Xylem

*Work done while at Cognizant

Abstract

The [Open *E. coli* Model](#) refers to a collection of models aimed to provide an estimation/classification of potential risk regarding *E. coli* levels in water bodies in the UK. They offer flexibility with regards to the water quality data that is available; there is an advanced version which uses water quality data and a light version which works solely on weather, satellite and land cover data.

1 INTRODUCTION

Monitoring microbial water quality is essential for protecting public health^[1]. *Escherichia coli* (*E. coli*) is a widely used indicator of faecal contamination in recreational and bathing waters, yet traditional testing methods require laboratory analysis that takes >1 working day to deliver results. This time lag means the public often learns of contamination risks too late. Our AI-driven Open *E. coli* Model uses machine learning to estimate *E. coli* concentrations in real time based on environmental, meteorological, and land-use data. Two complementary approaches were developed:

- *E. coli* Intensity Estimator – predicts microbial concentration levels (CFU/100ml).
- *E. coli* Classifier – categorises contamination risk using thresholds relevant to UK and international bathing water standards.

The models achieve accuracy levels of $\sim 80\%$ under validation testing, providing an early-warning tool that can inform swimmers, regulators, and water managers before risks escalate. By leveraging existing datasets—such as water quality archives, satellite imagery, and weather analysis—the system reduces reliance on costly and delayed laboratory results, while extending coverage across both monitored and unmonitored sites.

The Open *E. coli* Model demonstrates the potential of AI to transform water quality monitoring into a proactive, real-time public health safeguarding tool.

1.1 RIVER DEEP MOUNTAIN AI

[River Deep Mountain AI](#) is an innovation project funded by the [Ofwat Innovation Fund](#) working collaboratively to develop open-source AI/ML models that can inform effective actions to tackle waterbody pollution.

The project consists of 6 core partners: Northumbrian Water, Cognizant Ocean, Xylem Inc., Water Research Centre Limited, The Rivers Trust and ADAS. The project is further supported by 6 water companies across the United Kingdom and Ireland.

2 BACKGROUND AND CONTEXT

2.1 REGULATORY OR ENVIRONMENTAL IMPORTANCE

E. coli is the primary regulatory indicator for fecal contamination in surface waters. High levels indicate potential presence of pathogens harmful to human health^[2,3], prompting authorities to issue bathing advisories or close recreational waters. In England, more than 450 designated bathing waters require regular testing to meet Environment Agency (EA) water quality standards. These can be seen in [Table 1](#).

2.2 TRADITIONAL METHODS

Conventional monitoring relies on laboratory testing of water samples. This process requires specialised facilities and typically delays results by at least 18 hours.

2.3 MONITORING CHALLENGES

- **Sampling frequency:** Testing occurs periodically, and have limited temporal resolution.
- **Result delays:** Lab turnaround means users may swim in unsafe waters before contamination is reported.
- **Limited coverage:** Limited monitoring sites leave gaps in spatial understanding of water quality.
- **High costs:** Routine lab testing and sampling are resource intensive.

2.4 OPPORTUNITY FOR AI

AI offers a solution by using proxy indicators and environmental data to continuously estimate *E. coli* levels continuously, which can be used as a supportive tool. Machine learning can generate estimates based on weather, land use, and water quality patterns, provide real-time or near real-time risk alerts, and expand monitoring to untested locations through scalable, data-driven inference^[1,5].

Organization	Thresholds (CFU/100ml)	Classification
EA (coastal ^[4])	$\leq 250 = \text{Excellent}; \leq 500 = \text{Good}; > 500 = \text{Poor}$	Bathing water quality
EA (inland ^[4])	$\leq 500 = \text{Excellent}; \leq 1000 = \text{Good}; > 1000 = \text{Poor}$	Bathing water quality

Table 1: Environment Agency Regulatory Thresholds for *E. coli* in Recreational Waters

3 PROBLEM STATEMENT

Traditional methods cannot deliver timely or spatially comprehensive *E. coli* monitoring. The central question is whether *E. coli* levels or contamination risk can be estimated and classified at high frequency using already measured parameters (e.g., temperature, ammonia) along with environmental context (weather, land cover, satellite data). The Open *E. coli* Model addresses this challenge by providing machine learning tools that augment regulatory testing with faster, broader, and more cost-effective monitoring.

4 DATA SOURCES

The Open *E. coli* Model has been trained and validated using a combination of regulatory water quality data archive, satellite, and environmental datasets, covering England, Wales, and Scotland [6,7,8,9,10,11,12,13].

4.1 WATER QUALITY PARAMETERS

- **Environment Agency (EA) – Water Quality Archive (WIMS):** ~ 68 million samples from 2,582 locations (2000–mid-2024). $\sim 157,000$ samples included direct *E. coli* measurements for the light model; after filtering by proxy water quality parameters, 21,000 samples from 1,027 locations were used for the advanced model. Parameters included water temperature, ammonia, nitrate, nitrite, phosphate, and suspended solids (105°C), with correlation analyses shown in Fig 1.
- **Natural Resources Wales (NRW) and Scottish Environment Protection Agency (SEPA):** $\sim 39k$ samples (Scotland) and $\sim 24k$ samples (Wales), primarily used in the light model to maximise sample size without direct water quality proxies.

4.2 LAND COVER DATA

CORINE Land Cover dataset (2018 edition), classifying the UK into 44 land use categories. According to Fig. 2, Features derived within a 10 km buffer around each sampling point (e.g., proportion of agricultural land, urban fabric, wetlands).

4.3 SATELLITE DATA (MODIS)

Moderate Resolution Imaging Spectroradiometer (MODIS) with 8-day composites at $500m$ resolution^[14]. Derived features included are Leaf Area Index, Fraction

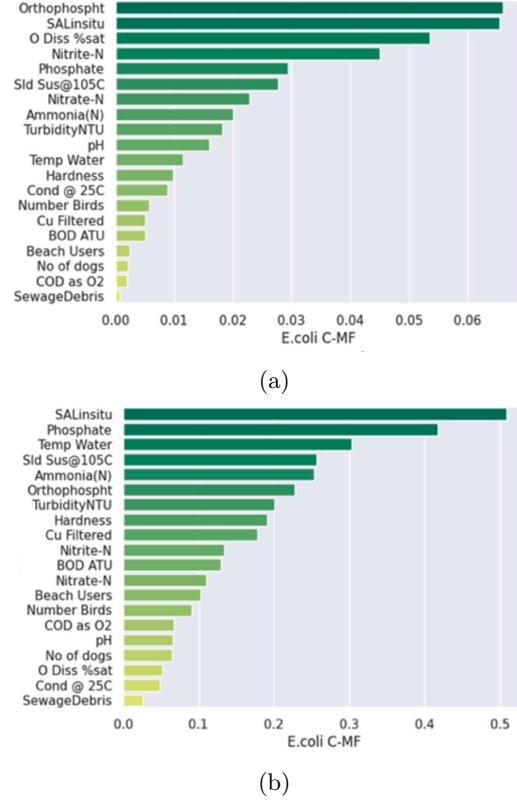


Figure 1: The Pearson (a) and Spearman (b) correlation of the water quality parameters.

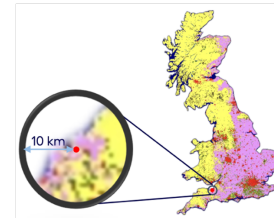


Figure 2: Land Cover Map of the UK

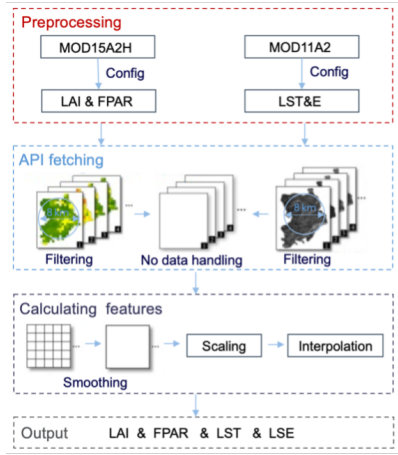


Figure 3: Block diagram of extraction of MODIS features

of Photosynthetically Active Radiation, and Land Surface Temperature, and Emissivity (see Fig. 3).

4.4 WEATHER DATA

Open-Meteo Historical Weather API was used to provide access to historical weather data, which includes 80 years of hourly data^[15,16]. 45 variables included (e.g., temperature, precipitation, wind, solar radiation, evapotranspiration). Aggregated at hourly, daily, and weekly scales.

4.5 PREPROCESSING STEPS

- Outlier removal using Z-score method (Appendix A & B in the report).
- Truncation of extreme *E. coli* values $> 10,000CFU/100ml$ to keep focus on regulatory thresholds (500 and $1000CFU/100ml$).
- Geographic buffer calculations for land cover.
- Alignment of weather and satellite time series to water sampling dates.

5 MODEL ARCHITECTURE

5.1 MODEL TYPES

Two machine learning approaches were developed:

- **Estimator (Regression):** Predicts *E. coli* concentration ($CFU/100ml$). Algorithms tested: XGBoost Regressor, LightGBM, Linear Regression, and Multi-Layer Perceptron (neural network).
- **Classifier (Binary Classification):** Predicts if *E. coli* levels exceed thresholds (500 or $1000CFU/100ml$). Algorithms tested: XGBoost Classifier, K-Nearest Neighbors, Support Vector Classifier.

Models were developed in **advanced** and **light** versions. The advanced model uses more features, including in-situ water quality, designed to achieve high performance at site-specific locations in England. The light model excludes water quality features but includes more samples from England, Wales, and Scotland, making it more generalizable and accessible.

5.2 FINAL SELECTION

XGBoost (XGB) outperformed other algorithms due to its strong performance on structured data, ability to handle non-linear relationships, and robustness to missing values and outliers.

5.3 INPUT FEATURES AND TARGETS

- **Advanced Model:** 65 features (water quality parameters + land cover + MODIS + weather).
- **Light Model:** 59 features (land cover + MODIS + weather; excluded water quality).
- **Estimator Target:** Continuous *E. coli* concentration ($CFU/100ml$).
- **Classifier Target:** Binary indicator of contamination risk ($\geq$ threshold).

6 TRAINING AND VALIDATION STRATEGY

6.1 DATA SPLITS

Two strategies were tested:

Random Temporal Split: 85% of samples for training, 15% for testing. Treats observations as independent based on low autocorrelation in *E. coli* measurements.

Random Geographic Split Training: Trained on most catchments (i.e. 861 for advanced dataset and 3385 for light dataset) and tested on the unseen catchments (i.e. 152 for advanced dataset and 598 for light dataset). This method is suitable when new unseen catchments are under study (see Fig. 4).

The temporal split produced stronger results ($\sim 80\%$ accuracy), but geographic split revealed reduced generalisation, highlighting site-dependency.

6.2 OVERFITTING PREVENTION

- Geographic split testing used to detect overfitting to local patterns.
- Z-score based outlier removal to avoid skew from extreme values.
- 10-fold cross validation

6.3 HYPERPARAMETER TUNING

Parameters tuned: *learning rate*, *max depth*, *n_estimators*, *subsample*, and *colsample_bytree*.

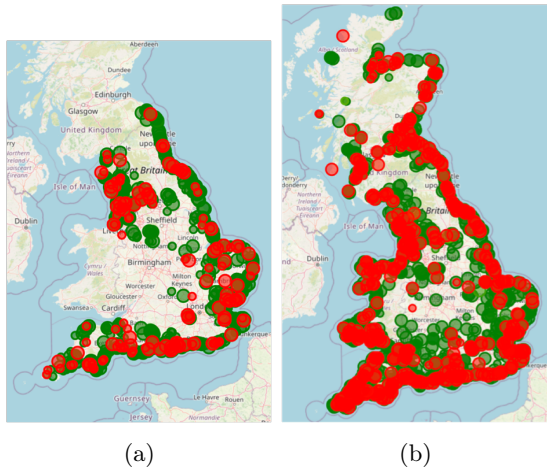


Figure 4: Catchment locations for the advanced (a) and the light (b) datasets. The green dots refer to the geographic split training locations and the red dots refer to test locations.

6.4 MODEL SCOPE

- **Advanced model:** Trained with fewer samples (approximately 21k samples), but more features (site-specific).
- **Light model:** Trained across EA, NRW, and SEPA datasets (approximately 215k samples), more generalisable but without in-situ water chemistry.

7 MODEL PERFORMANCE METRICS

The Open *E. coli* Model was evaluated under regression (Estimator) and classification (Classifier) frameworks. Regression performance was assessed using R^2 , RMSE, and MAE, and a confusion matrix was also constructed using thresholds of 500 and 1000CFU/100ml to explore the potential of the estimator as a classifier. Classification performance was measured using Accuracy, Precision, Recall, F1-Score, False Positives (FP), and False Negatives (FN).

7.1 OVERALL PERFORMANCE

Table 2 summarises the performance of different model versions across various data subsets. For the classifier, temporal split results show that light models achieve the highest accuracy (87.6%) and F1-score (0.86), although per-class performance is imbalanced, detecting lower *E. coli* concentrations more effectively. Advanced models achieve slightly lower temporal split accuracy (up to 81.7%) but more balanced regression performance. Geographic split evaluation reveals reduced accuracy due to site-specific patterns, highlighting some overfitting. Similar trends are observed for regression models: light regressors achieved up to 85.6% accuracy, whereas advanced regressors reached 81.7% accuracy. Attempts to augment high-concentration classes via sample duplication and additive noise did not improve performance.

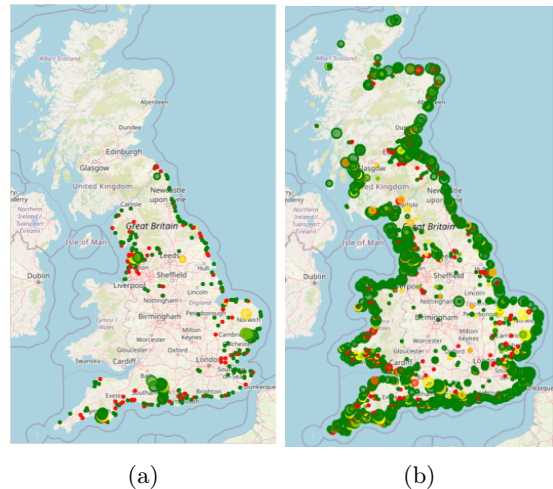


Figure 5: Maps showing the results for the advanced (a) and the light (b) test datasets for the classifiers with a temporal split with a threshold of 1000CFU/100ml. Green > 75%, Yellow > 65%, Orange > 55% and Red < 55% are the colors of the locations.

7.2 KEY OBSERVATIONS

- All final models perform significantly better than earlier baselines (version 1 through 3) see table 2.
- It can be observed that our light models achieve higher performance in comparison to advanced models. Although, it might not be a fair comparison, as light and advanced models have been tested on different datasets.
- Train datasets for both advanced and light models are unbalanced with low number of samples with high concentration of *E. coli*. This imbalance datasets was alleviated to some extent by reducing the threshold to 500CFU/100ml¹.
- Light classifier model achieved **highest classification accuracy** (87.6%) and F1 (0.86) over temporal split. Although, its per-class classification performance was imbalanced, and it performed better in detecting lower concentration of *E. coli*. On the other hand, light regressor model with accuracy of 85.6% provided somewhat balanced performance over both classes.
- The advanced regressor model achieved **highest classification accuracy** (81.7%) with a temporal split strategy.
- Drop in performance in geospatial split (see Section 5.1 in the report^[17]) indicates some overfitting to locations and/or existent of catchment specific characteristics.

¹We tried to augment the high concentration class by duplication these samples and adding low power additive noise, but no visible improvement in performance was seen.

E. coli Classifier	Random Temporal Split Training						Geographic Split Training					
Model_dataset_thr	Accuracy	Precision	Recall	F1-score	FP (Per class)	FN (Per class)	Accuracy	Precision	Recall	F1-score	FP (Per class)	FN (Per class)
Baseline 1.0: XGBC	72.75	0.72	0.73	0.72			70.41	0.68	0.70	0.69		
Baseline 2.0: XGBC	72.78	0.76	0.73	0.70	0.029	0.24	73.33	0.79	0.73	0.70	0.015	0.251
Baseline 3.0: XGBC	78.35	0.81	0.78	0.77	0.023	0.193	73.34	0.77	0.73	0.74	0.054	0.211
XGBCv4 adv 1000	80.77	0.81	0.80	0.80	0.057	0.137	75.29	0.80	0.75	0.76	0.052 (0.17)	0.194 (0.28)
XGBCv4 adv 500	80.86	0.80	0.80	0.80	0.094	0.102	80.36	0.82	0.80	0.81	0.073 (0.40)	0.123 (0.15)
XGBCv4 lgt 1000	87.61	0.87	0.87	0.86	0.030 (0.04)	0.095 (0.44)	80.07	0.77	0.80	0.77	0.049 (0.06)	0.145 (0.70)
XGBCv4 lgt 500	86.74	0.86	0.86	0.86	0.046 (0.06)	0.089 (0.30)	73.81	0.72	0.74	0.73	0.092 (0.14)	0.181 (0.56)

(a)

E. coli Estimator	Random Temporal Split Training						Geographic Split Training					
Model	Accuracy	MAE	R2	RMSE	FP (Per class)	FN (Per class)	Accuracy	MAE	RMSE	R2	FP (Per class)	FN (Per class)
Baseline 1.0: LGBMR	68.92	1267					64.13	1289				
Baseline 2.0: XGBR	70.66	1244	0.478	2001	0.228	0.065	71.83	1206	1999	0.494	0.189	0.092
Baseline 3.0: LGBMR	73.39	1138	0.556	1859	0.213	0.052	74.89	1595	2399	0.491	0.196	0.054
XGBRv4 adv 1000	81.72	984	0.557	1870	0.090	0.092	77.50	1883	2935	0.282	0.085	0.139
XGBRv4 adv 500	80.51	984	0.557	1870	0.127	0.067	83.49	1883	2935	0.282	0.099 (0.52)	0.056 (0.08)
XGBRv4 lgt 1000	85.61	845	0.497	1808	0.103	0.040	74.13	1227	2349	0.142	0.183 (0.22)	0.095 (0.41)
XGBRv4 lgt 500	82.48	845	0.497	1808	0.145	0.030	65.76	1227	2349	0.142	0.294 (0.39)	0.075 (0.24)

(b)

Table 2: Performance metrics of *E. coli* Classification (a) and Estimation (b) trained on different subsets of the data

- Low R^2 (0.142 for the light model), indicating very poor variance explanation.

8 DEPLOYMENT CONSIDERATIONS

The Open *E. coli* Model was developed primarily as an open source model, and the underlying architecture can support both retrospective analysis and near real-time deployment.

- 1. Retrospective Analysis:** The current implementation has been validated on recent datasets (EA, NRW, SEPA, MODIS, and Open-Meteo) from mid 2024 to mid 2025. This was useful for identifying long-term trends, site variability, and retrospective compliance checks.
- 2. Real-time Deployment Potential:**
 - With live weather feeds (e.g., Open-Meteo API) and MODIS data fetch, the classifier model could generate daily risk alerts.
 - Land cover is static, requiring only periodic refresh.
 - Integration with existing monitoring dashboards (e.g., Environment Agency bathing water portal) is feasible.
- 3. Required Inputs:**
 - MODIS and Weather variables (LAI, temperature, rainfall, solar radiation, evapotranspiration, wind).
 - Land cover data (proportion of agriculture, urban, wetland, etc.).
 - (Advanced model only) Selected water quality proxies such as ammonia, temperature, phosphate.

- 4. System Environment:** Models implemented in Python using XGBoost and scikit-learn. Training conducted on standard workstations; inference is lightweight and can run on cloud services or local servers. Prototype API endpoints could expose prediction results for external integration (not yet implemented in this study).

9 USE CASES AND APPLICATIONS

The Open *E. coli* Model offers flexible applications across regulatory, operational, and community domains:

1. Regulatory Compliance Support

- Provides early warning of exceedance against regulatory bathing water thresholds.
- Complements, rather than replaces, laboratory testing by extending coverage between sampling events.

2. Event Detection & Early Warning

- Detects short-term spikes in microbial risk linked to rainfall events, agricultural runoff, or sewer overflows.
- Helps authorities issue faster bathing advisories to protect public health.

3. Filling Gaps in Monitoring Coverage

- Extends microbial risk estimation to sites with sparse or no laboratory monitoring.

4. Citizen Science Verification

- Community groups can use model outputs to prioritise sites for field sampling.
- Acts as a “virtual sensor” network, augmenting limited physical infrastructure.

While promising, the Open *E. coli* Model has several limitations:

1. Generalisation

- Both the classification and estimation models suffer from significant performance drops in geographic splits (see Section 5.1 in the detailed model documentation^[17]), especially the light models.
- Although overall accuracy is high, imbalanced performance is observed, suggesting that most models fail to generalise to unseen validation datasets. Possible reasons include:
 - (a) Overfitting
 - (b) Data imbalance
 - (c) Missing important features
 - (d) High natural variability in *E. coli* levels
 - (e) Significant distribution shift in features
- Performance comparison on temporal and geographical splits indicates overfitting to known locations — a critical limitation for real-world deployment across diverse coastal areas. This is less of an issue in already designated bathing waters, but poses challenges in newly designated inland and coastal bathing waters.
- Higher accuracy in temporal splits suggests reliance on localised patterns, which do not translate well to geographical splits. The performance drop in geographical splits indicates strong location-dependence or overfitting to temporal patterns.
- **Light vs. Advanced Trade-off:** Light models (without in-situ water quality data) perform better in temporal splits, making them suitable for real-time, low-resource scenarios. Advanced models perform better in geographical splits, highlighting the importance of in-situ data when applying these models in unseen catchments.

2. False Negatives (FN) and False Positives (FP)

- FN and FP rates are considerably higher in geographical splits, especially for light models, indicating generalisation issues (see Section 5.1).
- Higher FN rates in light models raise concerns about under-detection of high-risk events, which is critical for public health.

FN is consistently higher than FP in most setups, as missing contaminated sites has greater public health implications.

3. Threshold Sensitivity

- Changing classification thresholds significantly affects the FN/FP balance.

4. Data Limitations

- Water quality data archives are biased toward coastal bathing waters, limiting inland generalisation.
- MODIS satellite data are coarse (500m, 8-day composites), limiting event-scale resolution.

5. Uncertainty Quantification

- The current implementation does not explicitly provide estimation intervals.
- Future development could incorporate:
 - Class balancing (e.g., resampling techniques)
 - Adjusting decision thresholds
 - Cost-sensitive training
 - Ensemble strategies (e.g., regression + classification pipeline)
 - Uncertainty estimation or confidence intervals — especially important due to natural noise in environmental data
 - Inclusion of more relevant features

Despite these limitations, the Open *E. coli* Model represents a significant step forward in scalable, near-real-time microbial water quality assessment, paving the way for integration into national monitoring strategies.

BIBLIOGRAPHY

- [1] Seok Min Hong, Billie J. Morgan, Matthew D. Stocker, Jaclyn E. Smith, Moon S. Kim, Kyung Hwa Cho, and Yakov A. Pachepsky. Using machine learning models to estimate escherichia coli concentration in an irrigation pond from water quality and drone-based rgb imagery data. *Water Research*, 260:121861, 2024. ISSN 0043-1354. doi: <https://doi.org/10.1016/j.watres.2024.121861>. URL <https://www.sciencedirect.com/science/article/pii/S0043135424007620>.
- [2] URL <https://www.health.state.mn.us/diseases/ecoli>.
- [3] Guodong Wang and Michael P. Doyle. Survival of enterohemorrhagic escherichia coli o157:h7 in water. *Journal of Food Protection*, 61(6):662–667, 1998. ISSN 0362-028X. doi: <https://doi.org/10.4315/0362-028X-61.6.662>. URL <https://www.sciencedirect.com/science/article/pii/S0362028X22015964>.
- [4] The bathing water regulations 2013. URL <https://www.legislation.gov.uk/uksi/2013/1675>.

- [5] Ai-driven system aims to inform public of immediate health risks from bacterial water pollution. URL <https://www.wpgw.org/ai-driven-system-aims-to-inform-public-of-immediate-health-risks-from-bacterial-water-pollution>.
- [6] Yael Rozen and Shimshon Belkin. Survival of enteric bacteria in seawater. *FEMS Microbiology Reviews*, 25(5):513–529, 12 2001. ISSN 0168-6445. doi: 10.1111/j.1574-6976.2001.tb00589.x. URL <https://doi.org/10.1111/j.1574-6976.2001.tb00589.x>.
- [7] Slaven Jozić and Mladen Solic. *Effect of Environmental Conditions on Escherichia coli Survival in Seawater*. 07 2017. ISBN 978-953-51-3329-2. doi: 10.5772/67912. URL https://www.researchgate.net/publication/318401939_Effect_of_Environmental_Conditions_on_Escherichia_coli_Survival_in_Seawater.
- [8] Stephen DeVilbiss, M. Steele, Leigh-Anne Krometis, and Brian Badgley. Freshwater salinization increases survival of escherichia coli and risk of bacterial impairment. *Water Research*, page 116812, 01 2021. doi: 10.1016/j.watres.2021.116812. URL <https://pubmed.ncbi.nlm.nih.gov/33461082/>.
- [9] Dania Albin, Lauren Lester, Philip Sanders, Jocelyne Hughes, and Michelle C. Jackson. The combined effects of treated sewage discharge and land use on rivers. *Global Change Biology*, 29(22):6415–6422, 2023. doi: <https://doi.org/10.1111/gcb.16934>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcb.16934>.
- [10] Maryam Taabodi, Fawzy Mohamed Hashem, Thomas Patrick Oscar, Salina Parveen, and Eric Bruce May. The Possible Roles of Escherichia coli in the Nitrogen Cycle. *International Journal of Environmental Research*, 13(3):597–602, June 2019. doi: 10.1007/s41742-019-00191-y. URL <https://link.springer.com/article/10.1007/s41742-019-00191-y>.
- [11] The effects of nutrients on the survival of escherichia-coli in lake water. *JOURNAL OF APPLIED BACTERIOLOGY*, 66(6):559–569, June 1989. ISSN 0021-8847. URL <https://wrap.warwick.ac.uk/id/eprint/23818/>.
- [12] Lixia Wang, Shangping Xu, and Jin Li. Effects of phosphate on the transport of escherichia coli o157:h7 in saturated quartz sand. *Environmental Science & Technology*, 45(22):9566–9573, 2011. doi: 10.1021/es201132s. URL <https://doi.org/10.1021/es201132s>. PMID: 21955132.
- [13] J. Jang, H.-G. Hur, M.J. Sadowsky, M.N. Byappanahalli, T. Yan, and S. Ishii. Environmental escherichia coli: ecology and public health implications—a review. *Journal of Applied Microbiology*, 123(3):570–581, 2017. doi: <https://doi.org/10.1111/jam.13468>. URL <https://enviromicro-journals.onlinelibrary.wiley.com/doi/abs/10.1111/jam.13468>.
- [14] Knyazikhin Y. Park T. Myneni, R. Modis/terra leaf area index/fpar 8-day l4 global 500m sin grid v061 [data set]. nasa land processes distributed active archive center, 2021. URL <https://www.earthdata.nasa.gov/data/catalog/lpcloud-mod15a2h-061>.
- [15] Open-meteo – open-source weather api. URL <https://open-meteo.com/>.
- [16] Open-meteo – historical weather api. URL <https://open-meteo.com/en/docs/historical-weather-api>.
- [17] Open e.coli models – rdmai, 2025. URL <https://github.com/Cogniznt-RDMAI/Open-Ecoli-Models>.

Disclaimer

This document has been prepared by the RDMAI consortium and is intended for informational purposes only.

The model described in this document is the product of an ongoing research and development project. It represents an early iteration and its outputs are preliminary. The model is not intended for use in production environments, regulatory reporting, environmental monitoring, or any decision-making in relation to public health, public policy, or commercial matters without independent validation and the acceptance of full liability by the user.

All use of the model and its associated software is subject to the full terms set out in the Open E.ColiModel repository README, available at <https://github.com/Cognizant-RDMAI/Open-Ecoli-Models>. Readers seeking technical detail on model performance, limitations, and data sources should refer to the accompanying Model Report. The RDMAI consortium and its members accept no liability for any outcomes arising from use of or reliance on the model or its outputs.